OR-AS Operations Research Applications and Solutions	Case Name: Residential House Finishing Works (4)	Sector	Construction (residential building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	William & Jef		User defined distributions
Date	2016	Project Control	Automatic tracking
File Name	C2016-19 Residential House Finishing Works (4)		Tracking based on user input

1. Project description

Project authenticity

Residential House Finishing Works (4) is a project focused on the phased completion of a residential building, including installation of ventilation, plumbing, electrical systems, insulation, and underfloor heating, followed by interior works such as flooring, carpentry, and kitchen installation, and concluding with final finishing to deliver a fully functional and ready-to-occupy home.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	30
Planned Duration (PD)	91d
Budget At Completion (BAC)	51,303.38 €
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	70%
Activity Distribution (AD)	70%
Length of Arcs (LA)	50%
Topological Float (TF)	43%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	100	0	N/A
CRI-tau	100	0	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	71	37	-0.87
SI	84	29	-2.10
SSI	17	11	0.33
CRI-r	17	13	1.08
CRI-rho	17	13	1.19
CRI-tau	10	8	1.47

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. *Project Control*

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	2.32%	1.30%
PV - SPI	4.46%	2.57%
PV - SCI	4.46%	2.57%
ED - 1	2.89%	1.77%
ED - SPI	4.43%	2.53%
ED - SCI	4.43%	2.53%
ES - 1	1.97%	-0.75%
ES - SPI(t)	8.07%	-5.86%
ES - SCI(t)	8.07%	-5.86%

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.00%	0.00%
CPI	0.00%	0.00%
SPI	2.19%	1.25%
SPI(t)	6.85%	-5.50%
SCI	2.19%	1.25%
SCI(t)	6.85%	-5.50%
0.8 CPI + 0.2 SPI	0.51%	0.37%
0.8 CPI + 0.2 SPI(t)	1.06%	-0.77%

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 4 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-1057.00	0.00	0.00	0.97	1.00	1.00	1.00
std dev	1064.95	0.00	0.00	0.02	0.00	0.00	0.00
final	-2048.00	0.00	0.00	0.96	1.00	1.00	1.00

2.3.3.2. Time forecasting

PD	91d	Real Duration	91d	Late/early	0%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	91.00	0.00	0.00	0.00
PV - SPI	91.00	0.00	0.00	0.00
PV - SCI	94.50	2.08	3.85	-3.85
ED - 1	91.00	0.00	0.00	0.00
ED - SPI	91.00	0.00	0.00	0.00
ED - SCI	91.75	0.96	0.82	-0.82
ES - 1	91.00	0.00	0.00	0.00
ES - SPI(t)	91.00	0.00	0.00	0.00
ES - SCI(t)	91.75	0.96	0.82	-0.82

2.3.3.3. Cost forecasting

BAC	51,303.38 €	Real Cost	53,351.38 €	Over/under budget	4%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	52360.38	1064.95	1.86	1.86
CPI	52932.46	1163.42	1.54	0.79
SPI	52360.38	1064.95	1.86	1.86
SPI(t)	52360.38	1064.95	1.86	1.86
SCI	52932.46	1163.42	1.54	0.79
SCI(t)	52932.46	1163.42	1.54	0.79
0.8 CPI + 0.2 SPI	52814.43	1112.61	1.58	1.01
0.8 CPI + 0.2 SPI(t)	52814.43	1112.61	1.58	1.01